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Reality Underground

It was the Summer of Love in New York City, and John Clauser was cooped up in a room at the Goddard Institute for Space Studies on 112th Street, teasing secrets from the oldest light in the universe. Clauser, a physics graduate student at Columbia, was attempting to measure the recently discovered cosmic microwave background (CMB) radiation, an echo of the Big Bang itself. It was difficult and painstaking work at the cutting edge of science—the CMB, a faint static hiss of radio across the sky, had been discovered only three years previously, by a pair of physicists working at Bell Labs. Only one other group of physicists had managed to detect it since. Clauser and his graduate advisor, Patrick Thaddeus, were intent on being the next to hear the beginning of the universe and to do it more accurately than any before them. But, one day in 1967, Clauser made a different kind of discovery altogether. Looking for the latest research in the Goddard Institute library, Clauser stumbled upon a journal with the unusual name *Physics Physique Fizika.* Intrigued, he flipped it open, and a paper caught his eye: “On the Einstein-Podolsky-Rosen Paradox,” by one J. S. Bell.

Clauser was young, brash, voluble—and had already been doubting the Copenhagen interpretation for years. Clauser’s father, Francis, had a PhD in aeronautics from Caltech (as did Francis’s twin brother Milton) and had trained John to be skeptical. “Son, look at the data,” Francis told him. “People will have lots of fancy theories, but always go back to the original data and see if you come to the same conclusions.…Common wisdom is